```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
```

```
df = pd.read_csv('/content/project.csv')
```

```
df
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	Educat
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Lif
1	49	No	Travel_Frequently	279	Research & Development	8	1	Lif
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Lif
4	27	No	Travel_Rarely	591	Research & Development	2	1	
...	...	...	...	...	...	...	...	
1465	36	No	Travel_Frequently	884	Research & Development	23	2	
1466	39	No	Travel_Rarely	613	Research & Development	6	1	
1467	27	No	Travel_Rarely	155	Research & Development	4	3	Lif
1468	49	No	Travel_Frequently	1023	Sales	2	3	
1469	34	No	Travel_Rarely	628	Research & Development	8	3	

1470 rows × 35 columns

```
df.shape
```

```
(1470, 35)
```

```
df.head()
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	Education
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sc
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sc
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sc
4	27	No	Travel_Rarely	591	Research & Development	2	1	M

5 rows × 35 columns

```
print("missing values in the dataset:")
print(df.isnull().sum())
```

```
missing values in the dataset:
Age                         0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
YearsWithCurrManager        0
dtype: int64
```

```
print("\nSummary statistics:")
print(df.describe())
```

```
Summary statistics:
               Age    DailyRate  DistanceFromHome    Education  EmployeeCount  \
count  1470.000000  1470.000000       1470.000000  1470.000000         1470.0
mean     36.923810   802.485714          9.192517     2.912925            1.0
std       9.135373   403.509100          8.106864     1.024165            0.0
min      18.000000   102.000000          1.000000     1.000000            1.0
25%      30.000000   465.000000          2.000000     2.000000            1.0
50%      36.000000   802.000000          7.000000     3.000000            1.0
75%      43.000000  1157.000000         14.000000     4.000000            1.0
max      60.000000  1499.000000         29.000000     5.000000            1.0

       EmployeeNumber  EnvironmentSatisfaction   HourlyRate  JobInvolvement  \
count     1470.000000              1470.000000  1470.000000     1470.000000
mean      1024.865306                 2.721769    65.891156        2.729932
std        602.024335                 1.093082    20.329428        0.711561
min          1.000000                 1.000000    30.000000        1.000000
25%        491.250000                 2.000000    48.000000        2.000000
50%       1020.500000                 3.000000    66.000000        3.000000
75%       1555.750000                 4.000000    83.750000        3.000000
max       2068.000000                 4.000000   100.000000        4.000000

          JobLevel  ...  RelationshipSatisfaction  StandardHours  \
count  1470.000000  ...               1470.000000         1470.0
mean      2.063946  ...                  2.712245           80.0
std       1.106940  ...                  1.081209            0.0
min       1.000000  ...                  1.000000           80.0
25%       1.000000  ...                  2.000000           80.0
50%       2.000000  ...                  3.000000           80.0
75%       3.000000  ...                  4.000000           80.0
max       5.000000  ...                  4.000000           80.0

       StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  \
count       1470.000000        1470.000000            1470.000000
mean           0.793878          11.279592               2.799320
std            0.852077           7.780782               1.289271
min            0.000000           0.000000               0.000000
25%            0.000000           6.000000               2.000000
50%            1.000000          10.000000               3.000000
75%            1.000000          15.000000               3.000000
max            3.000000          40.000000               6.000000

       WorkLifeBalance  YearsAtCompany  YearsInCurrentRole  \
count      1470.000000     1470.000000         1470.000000
mean          2.761224        7.008163            4.229252
std           0.706476        6.126525            3.623137
min           1.000000        0.000000            0.000000
25%           2.000000        3.000000            2.000000
50%           3.000000        5.000000            3.000000
75%           3.000000        9.000000            7.000000
max           4.000000       40.000000           18.000000

       YearsSinceLastPromotion  YearsWithCurrManager
count              1470.000000           1470.000000
mean                  2.187755              4.123129
std                   3.222430              3.568136
min                   0.000000              0.000000
25%                   0.000000              2.000000
```

```
df.dropna(inplace=True)
```

```
df['Attrition'] = df['Attrition'].map({'Yes': 1, 'No': 0})
```

```
df['Age_Group'] = pd.cut(df['Age'], bins=[20, 30, 40, 50, 60, 70], labels=['20-30', '30-40', '40-5
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 36 columns):
 #   Column                    Non-Null Count  Dtype   
---  ------                    --------------  -----   
 0   Age                       1470 non-null   int64   
 1   Attrition                 1470 non-null   int64   
 2   BusinessTravel            1470 non-null   object  
 3   DailyRate                 1470 non-null   int64   
 4   Department                1470 non-null   object  
 5   DistanceFromHome          1470 non-null   int64   
 6   Education                 1470 non-null   int64   
 7   EducationField            1470 non-null   object  
 8   EmployeeCount             1470 non-null   int64   
 9   EmployeeNumber            1470 non-null   int64   
 10  EnvironmentSatisfaction   1470 non-null   int64   
 11  Gender                    1470 non-null   object  
 12  HourlyRate                1470 non-null   int64   
 13  JobInvolvement            1470 non-null   int64   
 14  JobLevel                  1470 non-null   int64   
 15  JobRole                   1470 non-null   object  
 16  JobSatisfaction           1470 non-null   int64   
 17  MaritalStatus             1470 non-null   object  
 18  MonthlyIncome             1470 non-null   int64   
 19  MonthlyRate               1470 non-null   int64   
 20  NumCompaniesWorked        1470 non-null   int64   
 21  Over18                    1470 non-null   object  
 22  OverTime                  1470 non-null   object  
 23  PercentSalaryHike         1470 non-null   int64   
 24  PerformanceRating         1470 non-null   int64   
 25  RelationshipSatisfaction  1470 non-null   int64   
 26  StandardHours             1470 non-null   int64   
 27  StockOptionLevel          1470 non-null   int64   
 28  TotalWorkingYears         1470 non-null   int64   
 29  TrainingTimesLastYear     1470 non-null   int64   
 30  WorkLifeBalance           1470 non-null   int64   
 31  YearsAtCompany            1470 non-null   int64   
 32  YearsInCurrentRole        1470 non-null   int64   
 33  YearsSinceLastPromotion   1470 non-null   int64   
 34  YearsWithCurrManager      1470 non-null   int64   
 35  Age_Group                 1442 non-null   category
dtypes: category(1), int64(27), object(8)
memory usage: 403.7+ KB
```

```
attrition_rate = df['Attrition'].mean() * 100
print(f"Attrition Rate: {attrition_rate:.2f}%")
```

```
Attrition Rate: 16.12%
```

```
data = pd.DataFrame(df)
```

```
data
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	Educat
0	41	1	Travel_Rarely	1102	Sales	1	2	Lif
1	49	0	Travel_Frequently	279	Research & Development	8	1	Lif
2	37	1	Travel_Rarely	1373	Research & Development	2	2	
3	33	0	Travel_Frequently	1392	Research & Development	3	4	Lif
4	27	0	Travel_Rarely	591	Research & Development	2	1	
...	...	...	...	...	...	...	...	
1465	36	0	Travel_Frequently	884	Research & Development	23	2	
1466	39	0	Travel_Rarely	613	Research & Development	6	1	
1467	27	0	Travel_Rarely	155	Research & Development	4	3	Lif
1468	49	0	Travel_Frequently	1023	Sales	2	3	
1469	34	0	Travel_Rarely	628	Research & Development	8	3	

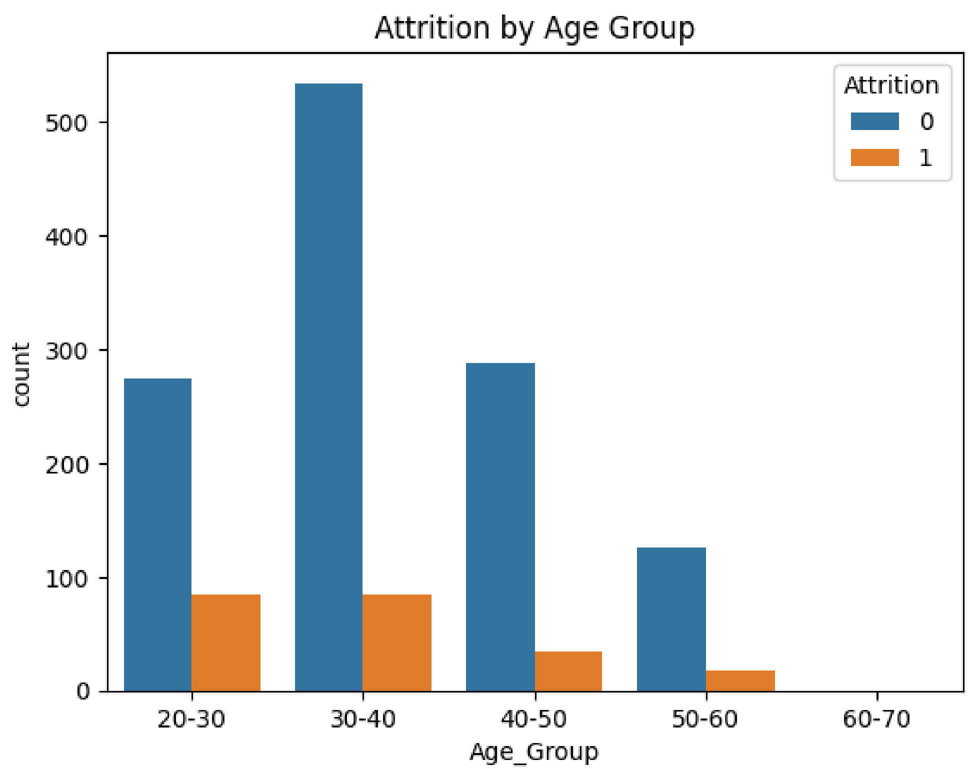
1470 rows × 36 columns

```
data.features_names = data.columns
```

```
<ipython-input-21-67edd8d1efae>:1: UserWarning: Pandas doesn't allow columns to be created via
  data.features_names = data.columns
```

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.countplot(x='Age_Group', hue='Attrition', data=df)
plt.title('Attrition by Age Group')
plt.show()
```




```
!pip install pydantic-settings
!pip install pandas-profiling
```

```
Collecting pydantic-settings
  Downloading pydantic_settings-2.6.1-py3-none-any.whl.metadata (3.5 kB)
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.10/dist-packages (fro
Collecting python-dotenv>=0.21.0 (from pydantic-settings)
  Downloading python_dotenv-1.0.1-py3-none-any.whl.metadata (23 kB)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.10/dist-packag
Requirement already satisfied: pydantic-core==2.23.4 in /usr/local/lib/python3.10/dist-package
Requirement already satisfied: typing-extensions>=4.6.1 in /usr/local/lib/python3.10/dist-pack
Downloading pydantic_settings-2.6.1-py3-none-any.whl (28 kB)
Downloading python_dotenv-1.0.1-py3-none-any.whl (19 kB)
Installing collected packages: python-dotenv, pydantic-settings
Successfully installed pydantic-settings-2.6.1 python-dotenv-1.0.1
Requirement already satisfied: pandas-profiling in /usr/local/lib/python3.10/dist-packages (3.
Requirement already satisfied: ydata-profiling in /usr/local/lib/python3.10/dist-packages (fro
Requirement already satisfied: scipy<1.14,>=1.4.1 in /usr/local/lib/python3.10/dist-packages (
Requirement already satisfied: pandas!=1.4.0,<3,>1.1 in /usr/local/lib/python3.10/dist-package
Requirement already satisfied: matplotlib<3.10,>=3.5 in /usr/local/lib/python3.10/dist-package
Requirement already satisfied: pydantic>=2 in /usr/local/lib/python3.10/dist-packages (from yd
Requirement already satisfied: PyYAML<6.1,>=5.0.0 in /usr/local/lib/python3.10/dist-packages (
Requirement already satisfied: jinja2<3.2,>=2.11.1 in /usr/local/lib/python3.10/dist-packages
Requirement already satisfied: visions<0.7.7,>=0.7.5 in /usr/local/lib/python3.10/dist-package
Requirement already satisfied: numpy<2.2,>=1.16.0 in /usr/local/lib/python3.10/dist-packages (
Requirement already satisfied: htmlmin==0.1.12 in /usr/local/lib/python3.10/dist-packages (fro
Requirement already satisfied: phik<0.13,>=0.11.1 in /usr/local/lib/python3.10/dist-packages (
Requirement already satisfied: requests<3,>=2.24.0 in /usr/local/lib/python3.10/dist-packages
Requirement already satisfied: tqdm<5,>=4.48.2 in /usr/local/lib/python3.10/dist-packages (fro
Requirement already satisfied: seaborn<0.14,>=0.10.1 in /usr/local/lib/python3.10/dist-package
Requirement already satisfied: multimethod<2,>=1.4 in /usr/local/lib/python3.10/dist-packages
Requirement already satisfied: statsmodels<1,>=0.13.2 in /usr/local/lib/python3.10/dist-packag
Requirement already satisfied: typeguard<5,>=3 in /usr/local/lib/python3.10/dist-packages (fro
Requirement already satisfied: imagehash==4.3.1 in /usr/local/lib/python3.10/dist-packages (fr
Requirement already satisfied: wordcloud>=1.9.3 in /usr/local/lib/python3.10/dist-packages (fr
Requirement already satisfied: dacite>=1.8 in /usr/local/lib/python3.10/dist-packages (from yd
```

```
    Requirement already satisfied: numba<1,>=0.56.0 in /usr/local/lib/python3.10/dist-packages (fr
    Requirement already satisfied: PyWavelets in /usr/local/lib/python3.10/dist-packages (from ima
    Requirement already satisfied: pillow in /usr/local/lib/python3.10/dist-packages (from imageha
    Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/dist-packages (fro
    Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.10/dist-packages (fr
    Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.10/dist-packages (from m
    Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.10/dist-packages (f
    Requirement already satisfied: kiwisolver>=1.0.1 in /usr/local/lib/python3.10/dist-packages (f
    Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (fro
    Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.10/dist-packages (fr
    Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.10/dist-packages
    Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.10/dist-pa
    Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-packages (from p
    Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.10/dist-packages (from
    Requirement already satisfied: joblib>=0.14.1 in /usr/local/lib/python3.10/dist-packages (from
    Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.10/dist-packag
    Requirement already satisfied: pydantic-core==2.23.4 in /usr/local/lib/python3.10/dist-package
    Requirement already satisfied: typing-extensions>=4.6.1 in /usr/local/lib/python3.10/dist-pack
    Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-pack
    Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from r
    Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (
    Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (
    Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.10/dist-packages (from s
    Requirement already satisfied: attrs>=19.3.0 in /usr/local/lib/python3.10/dist-packages (from
    Requirement already satisfied: networkx>=2.4 in /usr/local/lib/python3.10/dist-packages (from
```

```
from ydata_profiling import ProfileReport
```

```
ProfileReport(data)
```

```
Summarize dataset: 100%                                270/270 [01:09<00:00, 1.40it/s, Completed]
Generate report structure: 100%                        1/1 [00:25<00:00, 25.52s/it]
Render HTML: 100%                                      1/1 [00:09<00:00, 9.43s/it]
```

Pandas Profiling Report

Overview

Brought to you by YData

Overview | Alerts 27 | Reproduction

Dataset statistics

Number of variables	36
Number of observations	1470
Missing cells	28
Missing cells (%)	0.1%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	403.7 KiB
Average record size in memory	281.2 B

Variable types

Numeric	15
Categorical	19
Boolean	2

Variables

Select Columns

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 36 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   int64
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
```

```
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64
 19  MonthlyRate               1470 non-null   int64
 20  NumCompaniesWorked        1470 non-null   int64
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64
 24  PerformanceRating         1470 non-null   int64
 25  RelationshipSatisfaction  1470 non-null   int64
 26  StandardHours             1470 non-null   int64
 27  StockOptionLevel          1470 non-null   int64
 28  TotalWorkingYears         1470 non-null   int64
 29  TrainingTimesLastYear     1470 non-null   int64
 30  WorkLifeBalance           1470 non-null   int64
 31  YearsAtCompany            1470 non-null   int64
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
 35  Age_Group                 1442 non-null   category
dtypes: category(1), int64(27), object(8)
memory usage: 403.7+ KB
```

```
data['Attrition']
```

	Attrition
0	1
1	0
2	1
3	0
4	0
...	...
1465	0
1466	0
1467	0
1468	0
1469	0

1470 rows × 1 columns

dtype: int64

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import OneHotEncoder
```

```
categorical_features = ['BusinessTravel', 'Department', 'EducationField', 'Gender', 'JobRole', 'Ma

encoder = OneHotEncoder(sparse_output=False, handle_unknown='ignore')

encoded_features = encoder.fit_transform(data[categorical_features])

encoded_df = pd.DataFrame(encoded_features, columns=encoder.get_feature_names_out(categorical_feat

data = data.drop(columns=categorical_features)
data = pd.concat([data, encoded_df], axis=1)

x = data.drop(columns=['Attrition'])
y = data['Attrition']

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)

model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/linear_model/_logistic.py:469: ConvergenceWarn
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
LogisticRegression

LogisticRegression(max_iter=1000)
```

```
data.columns
```

```
Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
       'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
       'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate',
       'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',
       'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked',
       'Over18', 'OverTime', 'PercentSalaryHike', 'PerformanceRating',
       'RelationshipSatisfaction', 'StandardHours', 'StockOptionLevel',
       'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance',
       'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion',
```

```
           'YearsWithCurrManager', 'Age_Group'],
          dtype='object')
```

```
x=data[['Age', 'BusinessTravel', 'DailyRate', 'Department',
       'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
       'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate',
       'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',
       'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked',
       'Over18', 'OverTime', 'PercentSalaryHike', 'PerformanceRating',
       'RelationshipSatisfaction', 'StandardHours', 'StockOptionLevel',
       'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance',
       'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion',
       'YearsWithCurrManager', 'Age_Group']]
```

```
y=data[[ 'Attrition']]
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
```

```
x_train,x_test,y_train,y_test = train_test_split(x,y)
```

```
model.fit(x_train,y_train)
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/linear_model/_logistic.py:469: ConvergenceWarn
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_predict = model.predict(x_test)
```

```
print(y_predict)
```

```
[0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
 1 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 1
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0]
```

```
print(accuracy_score(y_test,y_predict))
```

```
0.8639455782312925
```